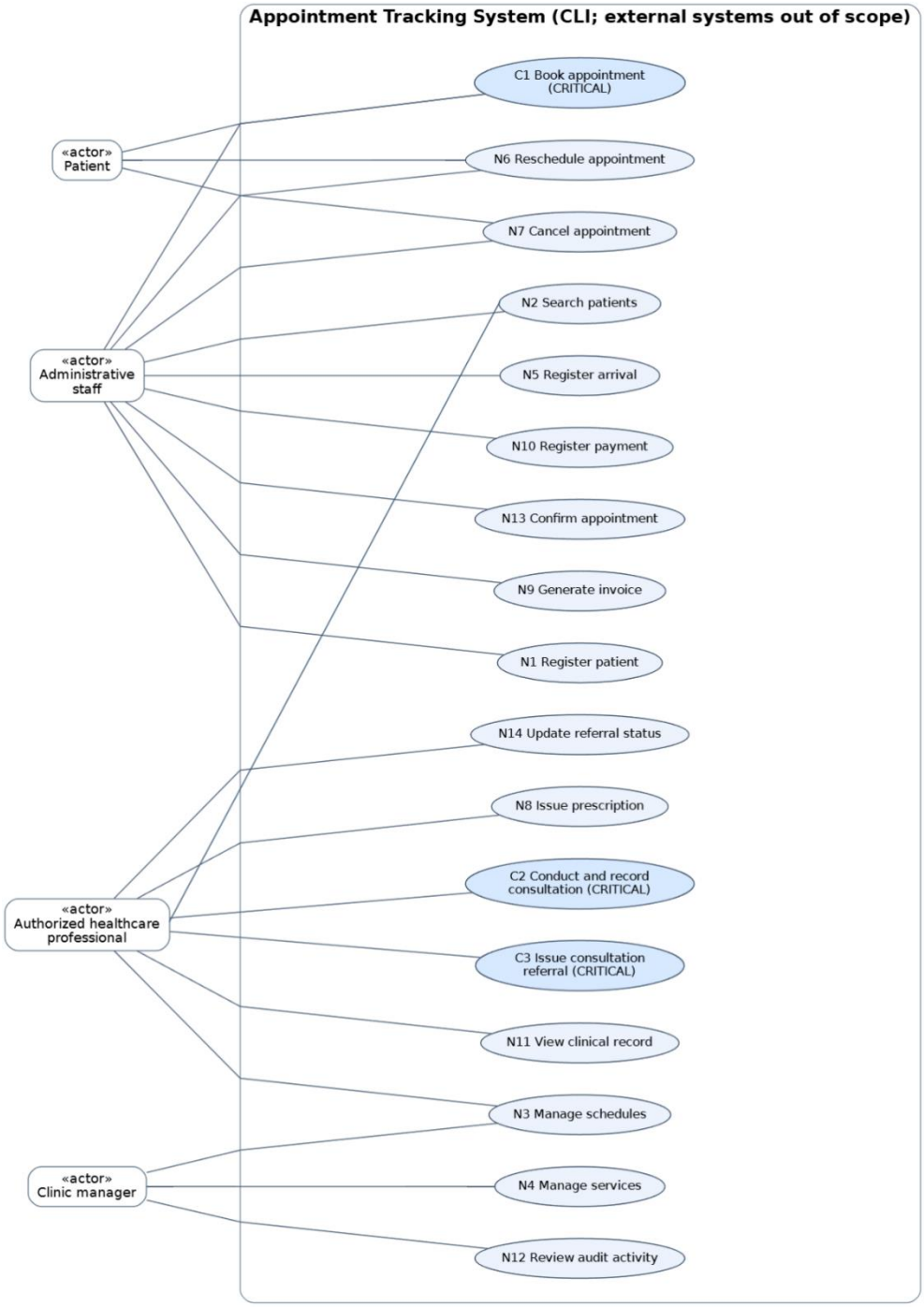
SOEN 342 - Project Iteration 1: Appointment Tracking System

Sarvesh Sai Rajesh – 40231819

Alex Khoury – 40299086

NIcholas Nguyen – 40317237


Appointment Tracking System (CLI; external systems out of scope)
C1 Book appointment (CRITICAL)
N6 Reschedule appointment
N7 Cancel appointment
N2 Search patients
N5 Register arrival
N10 Register payment
N13 Confirm appointment
N9 Generate invoice
N1 Register patient
N14 Update referral status
N8 Issue prescription
C2 Conduct and record consultation (CRITICAL)
C3 Issue consultation referral (CRITICAL)
N11 View clinical record
N3 Manage schedules
N4 Manage services
N12 Review audit activity
«actor» Patient
«actor» Administrative staff
«actor» Authorized healthcare professional
«actor» Clinic manager

Fully Dressed Critical Use-Case scenarios

C1 - Book Appointment (patient present at the clinic)

Primary actor: Administrative staff.

Supporting/interested actor: Patient (requests the appointment, normally through the staff member in this scenario).

Scope: Appointment Tracking System (black-box system).

Level: User goal.

Stakeholders and interests: Patient: obtain an appropriate appointment; administrative staff: register the correct booking efficiently; healthcare professional: avoid scheduling conflicts; clinic: enforce scheduling/referral requirements and protect patient information.

Preconditions: Administrative staff are identified, authenticated and authorized to manage appointments. The clinic offers appointment services and stores schedules. A matching patient record can be selected during the scenario.

Success guarantee (postconditions): A new Appointment is recorded with a unique identifier, scheduled interval and CONFIRMED status; it is associated with exactly one Patient, HealthcareProfessional, ClinicLocation and Service, and with an eligible Referral if required. Relevant booking activity is recorded for accountability.

Minimal guarantee (failure): An invalid booking creates no Appointment; existing appointments and schedules are preserved.

Main Success Scenario:

1. A patient present at the clinic asks administrative staff to arrange an appointment for a service.

2. The staff member submits patient search criteria to the system.

3. The system returns matching patients. The staff member selects the correct patient.

4. The staff member specifies the medical service, preferred professional, clinic location and desired time period.

5. The system returns candidate appointment time slots consistent with known schedule constraints.

6. The staff member chooses a slot and provides a referral reference if the service requires one.

7. The staff member requests the booking. The system rechecks current professional availability, applicable scheduling rules and referral eligibility.

8. The system creates and confirms the appointment, records its associations, and returns the appointment identifier and scheduled time.

Extensions (or alternative flows):

1. No matching patient: the system returns no matches. The staff member may arrange registration through N1, then restart booking.
2. No candidate slots: the system returns an empty set. The staff member may choose another period or professional.
3. The chosen slot is unavailable at final validation, such as after a concurrent booking. The system responds SLOT_UNAVAILABLE; no Appointment is created. The staff member may return to Step 4.
4. The service requires a referral, but an acceptable referral is missing or invalid. The system responds REFERRAL_REQUIRED or INVALID_REFERRAL; no Appointment is created. The staff member may obtain a referral or choose an eligible service.
5. The actor lacks booking authorization: the request is denied and patient information is not disclosed beyond that user's permissions.

Special Requirements: Access to clinical information is role-restricted. Patient lookups must not expose clinical records to administrative staff. Relevant booking activity must support accountability. The use case is technology-independent; the final application is CLI-based.

Technology and Data Variations: Patient searches can use insurance number (where present), date of birth, phone number, or partial first and/or last name. The chosen time period, professional, clinic location, service and optional referral are passed as input values.

When the patient requests the appointment directly, it is recorded with status REQUESTED and confirmed later through N13.

C2 - Conduct and Record Consultation

Primary actor: Authorized healthcare professional.

Supporting/interested actor: Patient (receives the consultation); administrative staff (registered the patient's arrival through N5, and may later book a follow-up appointment through C1).

Scope: Appointment Tracking System (black-box system).

Level: User goal.

Stakeholders and interests**:** Patient: receive appropriate care and have the consultation recorded accurately and confidentially; healthcare professional: record clinical findings quickly and against the correct appointment; administrative staff: know that the appointment was completed and whether a follow-up is needed, without access to clinical content; clinic: maintain complete clinical records, protect the confidentiality of clinical information and support accountability.

Preconditions: The healthcare professional is identified, authenticated and authorized to record clinical information. The patient is registered, has a clinical record, and has a scheduled appointment with this professional on the current day.

Success guarantee (postconditions): A new Consultation is recorded with a unique identifier, the date and time it was conducted, the observations, diagnosis, treatment recommendations and follow-up indicator. It is associated with exactly one Appointment and with the patient's ClinicalRecord. The appointment status is changed to COMPLETED. Relevant consultation activity is recorded for accountability. The consultation identifier is returned and can later be supplied when issuing a prescription (N8).

Minimal guarantee (failure): An invalid request creates no Consultation; the appointment status and the patient's clinical record are unchanged; clinical information is not disclosed beyond the user's permissions.

Main Success Scenario:

1. On the day of the appointment, after administrative staff have registered the patient's arrival (N5), the healthcare professional is ready to see the patient.
2. The professional requests the list of their appointments for the current day for which the patient has arrived.
3. The system returns the arrived appointments with the patient identifiers and permitted demographic details. The professional selects the appointment of the patient being seen.
4. The professional conducts the consultation with the patient.

5. The professional enters the clinical information (observations, diagnosis, treatment recommendations, and whether a follow-up is required) and requests that the consultation be recorded.
6. The system verifies authorization and validates the request: the appointment exists, is assigned to this professional, has status ARRIVED and has no consultation recorded yet, and the required clinical information is provided.
7. The system creates the Consultation with a unique identifier and the current date and time, associates it with the appointment and with the patient's clinical record, changes the appointment status to COMPLETED, and records the activity.
8. The system returns the consultation identifier.

Extensions (or alternative flows):

1. No arrived appointments (step 3): the system returns an empty list. The professional may wait until administrative staff register the patient's arrival through N5, then restart from step 2.
2. Patient has not arrived (step 6): the appointment status is still CONFIRMED because arrival was not registered. The system responds PATIENT_NOT_ARRIVED; no Consultation is created and the appointment is unchanged. Administrative staff may register the arrival through N5, and the professional resubmits from step 5.
3. Appointment not eligible (step 6): the appointment does not exist, or it is CANCELLED or already COMPLETED. The system responds APPOINTMENT_NOT_FOUND or INVALID_APPOINTMENT_STATUS; no Consultation is created and existing data is preserved.
4. Required information missing or invalid (step 6): the observations are missing or the follow-up indicator is not given. The system responds INVALID_CLINICAL_DATA; no Consultation is created. The professional may correct the data and resubmit from step 5.
5. The actor lacks authorization (step 2 or step 6), or the appointment is assigned to another professional: the request is denied with NOT_AUTHORIZED, no Consultation is created, and no clinical information is disclosed beyond that user's permissions.
6. Patient history needed (step 4): the professional may consult the patient's previous clinical information through N11, then continue from step 5.
7. Medication needed (after step 8): the professional issues a prescription through N8, using the returned consultation identifier.
8. Follow-up required (after step 8): the follow-up indicator is set, and administrative staff book the follow-up appointment through C1.

9. Specialist consultation needed (after step 8): the professional issues a consultation referral through C3.

Special Requirements: Observations, diagnosis and treatment recommendations are clinical information: they are visible only to authorized healthcare professionals. Administrative staff may see only that the appointment is COMPLETED and whether a follow-up is required. Recording a consultation neither issues a prescription nor books or reserves an appointment. Relevant consultation activity must support accountability; audit events reference the consultation by identifier and never reproduce clinical content. The use case is technology-independent; the final application is CLI-based.

Technology and Data Variations: The appointment is selected by its identifier from the list of arrived appointments. Observations, diagnosis and treatment recommendations are entered as text; the follow-up indicator is yes/no. The date and time of the consultation are set by the system. Diagnosis coding systems are not required at this phase.

C3 - Issue Consultation Referral

Primary actor: Authorized healthcare professional (referring professional).

Supporting/interested actor: Patient (needs the consultation); receiving specialist or professional; administrative staff (later book the resulting appointment); external referring physician (alternative flow 4, acting through authorized clinic staff).

Scope: Appointment Tracking System (black-box system).

Level: User goal.

Stakeholders and interests: Patient: be referred promptly to the right specialty with an appropriate priority; referring professional: record the referral accurately and quickly; receiving professional: receive a clear clinical reason and priority; administrative staff: obtain a valid referral reference for booking; clinic: enforce referral requirements and protect the confidentiality of clinical information.

Preconditions: The healthcare professional (or authorized staff member, in the external-referrer flow) is identified, authenticated and authorized to issue or register referrals. The clinic maintains its list of medical specialties and the professionals who hold them. A matching patient record can be selected during the scenario.

Success guarantee (postconditions): A new Referral is recorded with a unique identifier, the clinical reason, the priority and status ISSUED. It is associated with exactly one Patient, one referring professional (internal, or external with recorded details) and the requested Specialty, and with the requested HealthcareProfessional if one was specified. The referral

is not yet linked to any Appointment. Relevant referral activity is recorded for accountability. The referral identifier is returned and can later be supplied when booking (C1).

Minimal guarantee (failure): An invalid request creates no Referral; existing patients, referrals and appointments are preserved; clinical information is not disclosed beyond the user's permissions.

Main Success Scenario:

1. A healthcare professional decides that a patient needs a consultation with another professional or specialty and starts a consultation referral.
2. The professional submits patient search criteria to the system.
3. The system returns matching patients. The professional selects the correct patient.
4. The professional specifies the requested specialty (and optionally a particular professional), the clinical reason and the priority.
5. The professional requests that the referral be issued. The system verifies authorization and validates the data: the requested specialty exists, a named professional holds that specialty, the clinical reason is provided and the priority is one of the allowed values.
6. The system creates the Referral with a unique identifier and status ISSUED, associates it with the patient, the referring professional, the requested specialty and (if given) the requested professional, and records the activity.
7. The system returns the referral identifier and its status. The referral identifier can later be given to administrative staff to book the resulting appointment(s) through C1.

Extensions (or alternative flows):

1. No matching patient (step 3): the system returns no matches. The staff or professional may arrange registration through N1, then restart the referral.
2. Required information missing or invalid (step 5): the clinical reason is missing, the priority is not allowed, the requested specialty is unknown, or the named professional does not hold the requested specialty. The system responds INVALID_REFERRAL_DATA; no Referral is created. The professional may correct the data and resubmit from step 4.
3. The actor lacks authorization (step 2 or step 5): the request is denied with NOT_AUTHORIZED, no Referral is created, and no patient information is disclosed beyond that user's permissions.
4. External referring physician (step 4): the referring physician does not practise in the clinic. Authorized clinic staff register the referral on their behalf, supplying the

external physician's name, licence number and organization in place of an internal professional. The system records these details with the Referral, and the scenario continues from step 5.

Special Requirements: The clinical reason is clinical information: it is visible only to authorized healthcare professionals. Administrative staff may see only the referral identifier, requested specialty, priority and status. Issuing a referral neither books nor reserves an appointment. Relevant referral activity must support accountability. The use case is technology-independent; the final application is CLI-based.

Technology and Data Variations: Patient searches can use insurance number (where present), date of birth, phone number, or partial first and/or last name. Priority values: ROUTINE, URGENT, EMERGENT. Referral status values: ISSUED, COMPLETED, CANCELLED. The referrer is either an internal professional identifier or external physician details; the requested professional is optional.

Non-critical Use-Case scenario

N1 - Register Patient

Primary actor: Administrative staff

Trigger: A patient who is not yet registered requires a record in the clinic system.

Main success scenario: 1. An authorized staff member requests registration of a patient and enters identifying/demographic information. 2. The system checks the supplied information and verifies the uniqueness of the internal patient identifier. 3. The system creates the Patient record with a unique patient identifier and an empty clinical record; insurance information, if present, may be associated with the patient. 4. The system confirms the new patient identifier.

Failure scenario: If required identifying data is missing or invalid, the system rejects the registration and creates no Patient record.

System operation: registerPatient(patientData)

N2 - Search Patients

Primary actor: Administrative staff or Authorized healthcare professional

Trigger: An authorized clinic user needs to identify an existing patient.

Main success scenario: 1. An authorized clinic user supplies one or more permitted search criteria: insurance number, date of birth, phone number, or partial first/last name. 2. The system searches registered Patient records. 3. The system returns matching patient identifiers and permitted demographic details. 4. The authorized clinic user identifies the intended patient from the results.

Failure scenario: If there are no matches, the system returns an empty result (it does not register a patient automatically).

System operation: searchPatients(criteria)

Note: Patient search must not expose clinical notes to administrative staff.

N3 - Manage Schedules

Primary actor: Authorized healthcare professional or clinic manager

Trigger: Working availability must be added or modified for a professional at a clinic location.

Main success scenario: 1. An authorized actor identifies the professional and clinic location. 2. The actor submits available periods for that professional at that location. 3. The system checks scheduling constraints for the submitted periods. 4. The system records the accepted schedule and confirms the change.

Failure scenario: If any proposed period conflicts with an applicable scheduling constraint, the system rejects the change and retains the prior schedule.

System operation: updateProfessionalSchedule(professionalId, locationId, slots)

Note: The assignment requires schedule and constraint checking but does not specify every scheduling rule.

N4 - Manage Services

Primary actor: Clinic manager

Trigger: A medical service offered by the clinic must be defined or updated.

Main success scenario: 1. An authorized clinic manager selects a service to maintain or asks to define a new one. 2. The manager supplies the service information, including information needed to determine applicable appointment constraints and possible charges. 3. The system checks the proposed information. 4. The system stores the new or updated service definition and confirms the change.

Failure scenario: If essential service data is missing or invalid, the system rejects the change and leaves existing service information unchanged.

System operation: maintainService(serviceData)

N5 - Register Arrival

Primary actor: Administrative staff

Trigger: A patient arrives for an existing appointment on its scheduled day.

Main success scenario: 1. Staff identify the scheduled appointment for the arriving patient. 2. Staff ask the system to register the patient's arrival. 3. The system verifies that the appointment exists and is eligible for arrival registration. 4. The system records the arrival time against the appointment, changes its status to ARRIVED, and confirms check-in.

Failure scenario: If the appointment is unknown or cancelled, the system rejects check-in and creates no arrival record.

System operation: registerArrival(appointmentId)

N6 - Reschedule Appointment

Primary actor: Administrative staff (on behalf of the patient)

Trigger: A patient asks to move an existing confirmed appointment to a different time.

Main success scenario: 1. Staff identify the existing appointment and ask the system for alternative times with the same professional, location and service. 2. The system returns candidate slots consistent with known schedule constraints. 3. Staff select a new slot and request the reschedule. 4. The system verifies that the appointment exists, is CONFIRMED and has not yet taken place, then rechecks professional availability and applicable scheduling rules for the new slot. 5. The system replaces the appointment interval with the

new one, keeps the status CONFIRMED, records the activity, and returns the appointment identifier and new scheduled time.

Failure scenario: If the new slot is no longer available, or the appointment is unknown, cancelled or completed, the system rejects the request and the original appointment remains unchanged.

System operation: rescheduleAppointment(appointmentId, newStartsAt, newEndsAt)

Note: Step 1 reuses findAvailableSlots from C1. The original time is released only once the new time has been accepted, so the patient is never left without a slot. Any associated referral stays associated with the appointment.

N7 - Cancel Appointment

Primary actor: Administrative staff (on behalf of the patient)

Trigger: A patient (or the clinic) needs to cancel an existing appointment.

Main success scenario: 1. Staff identify the appointment and request its cancellation, optionally giving a reason. 2. The system verifies that the appointment exists, is REQUESTED or CONFIRMED, and has not yet taken place. 3. The system sets the appointment status to CANCELLED and releases the time slot in the professional's schedule. 4. The system records the activity and confirms the cancellation.

Failure scenario: If the appointment is unknown, already cancelled, or already completed, the system rejects the request and changes nothing.

System operation: cancelAppointment(appointmentId, reason?)

Note: Cancelling an appointment does not cancel or consume the associated referral; the referral stays ISSUED and can be used to book a new appointment. A CANCELLED appointment is also rejected by registerArrival (N5).

N8 - Issue Prescription

Primary actor: Authorized healthcare professional

Trigger: During or after a consultation, the professional decides that one or more medications should be prescribed.

Main success scenario: 1. The professional asks to issue a prescription linked to a consultation they conducted. 2. The professional supplies one or more medications, each with dosage, frequency, duration and instructions. 3. The system verifies that the consultation exists and was conducted by this professional, and that each medication is a known medication. 4. The system creates the Prescription with a unique identifier, associates it with the Consultation (and through it with the Patient and the professional), records each prescribed medication, and records the activity.

Failure scenario: If a medication is unknown, required prescription details are missing, or the consultation is unknown or belongs to another professional, the system rejects the request and creates no Prescription.

System operation: issuePrescription(consultationId, prescriptionItems)

Note: Transmission of the prescription to a pharmacy is external to the system at this phase. Only authorized healthcare professionals may issue prescriptions.

N9 - Generate Invoice

Primary actor: Administrative staff

Trigger: A patient has received an uninsured (paid) service and must be billed.

Main success scenario: 1. Staff identify the completed appointment and request an invoice for it. 2. The system verifies that the appointment exists, has status COMPLETED, that its service is not covered by RAMQ, and that no invoice already exists for it. 3. The system creates the Invoice with a unique identifier, the current date and an amount equal to the fee of the appointment's service, associates it with the appointment, and records the activity. 4. The system returns the invoice identifier and the amount due.

Failure scenario: If the appointment is unknown or not COMPLETED, its service is covered by RAMQ, or an invoice already exists for it, the system rejects the request and creates no Invoice.

System operation: generateInvoice(appointmentId)

Note: Electronic claims submission to RAMQ is external to the system at this phase. The invoice contains no clinical information.

N10 - Register Payment

Primary actor: Administrative staff

Trigger: A patient (or their private insurer) pays all or part of an invoice.

Main success scenario: 1. Staff identify the invoice and enter the amount paid and the payment method (for example cash, debit, credit card or private insurance). 2. The system verifies that the invoice exists, that the amount is greater than zero and does not exceed the remaining balance, and that a payment method is given. 3. The system creates the Payment with the amount, the current date and time and the payment method, associates it with the invoice, and records the activity. 4. The system returns the remaining balance of the invoice.

Failure scenario: If the invoice is unknown, the amount is zero, negative or greater than the remaining balance, or the payment method is missing, the system rejects the request and creates no Payment.

System operation: registerPayment(invoiceId, amount, method)

Note: An invoice may be settled by several partial payments. The remaining balance is the invoice amount minus the sum of its payments. Card processing and insurer reimbursement are external to the system.

N11 - View Clinical Record

Primary actor: Authorized healthcare professional

Trigger: A professional needs a patient's previous clinical information, for example before or during a consultation (C2).

Main success scenario: 1. The professional identifies the patient (through N2) and requests the patient's clinical record. 2. The system verifies that the actor is an authorized healthcare professional and that the patient exists. 3. The system returns the patient's consultations in chronological order (date, professional, observations, diagnosis, treatment recommendations and follow-up indicator) with their prescriptions. 4. The system records the access as an audit event.

Failure scenario: If the actor is not an authorized healthcare professional, the system rejects the request with NOT_AUTHORIZED and discloses no clinical information. If the patient is unknown, the system responds PATIENT_NOT_FOUND. If the patient has no consultations yet, the system returns an empty record.

System operation: viewClinicalRecord(patientId)

Note: This use case is read-only. Administrative staff never have access to clinical records. Every access is recorded so it can be reviewed through N12.

N12 - Review Audit Activity

Primary actor: Clinic manager

Trigger: The clinic needs to review who accessed or changed information, for example to investigate a confidentiality concern.

Main success scenario: 1. An authorized clinic manager supplies optional filters: user, patient, type of activity, and time period. 2. The system verifies that the manager is authorized to review audit information. 3. The system returns the matching audit events (user, role, type of activity, identifier of the affected record, date and time) in chronological order. 4. The manager reviews the results.

Failure scenario: If the actor is not authorized, the system rejects the request and discloses no audit data. If no events match, the system returns an empty result.

System operation: reviewAuditEvents(filter)

Note: Audit events reference records by identifier and never reproduce clinical content. The review itself is recorded as an audit event.

N13 - Confirm Appointment

Primary actor: Administrative staff

Trigger: A patient has requested an appointment directly, and the request is waiting for confirmation by the clinic.

Main success scenario: 1. Staff identify a requested appointment and ask the system to confirm it. 2. The system verifies that the appointment exists and has status REQUESTED, then rechecks professional availability, applicable scheduling rules and referral eligibility,

as in C1. 3. The system changes the appointment status to CONFIRMED and records the activity. 4. The system returns the appointment identifier and its scheduled time.

Failure scenario: If the appointment is unknown or its status is not REQUESTED, the system rejects the request and changes nothing. If the slot is no longer available or a required referral is missing or invalid, the system rejects the confirmation and the appointment remains REQUESTED.

System operation: confirmAppointment(appointmentId)

Note: Appointments requested directly by a patient through C1 are recorded with status REQUESTED; appointments booked by administrative staff are confirmed immediately by C1. The time slot stays reserved while the appointment is REQUESTED.

N14 - Update Referral Status

Primary actor: Authorized healthcare professional

Trigger: The outcome of a referral is known (the requested consultation took place) or the referral is no longer needed.

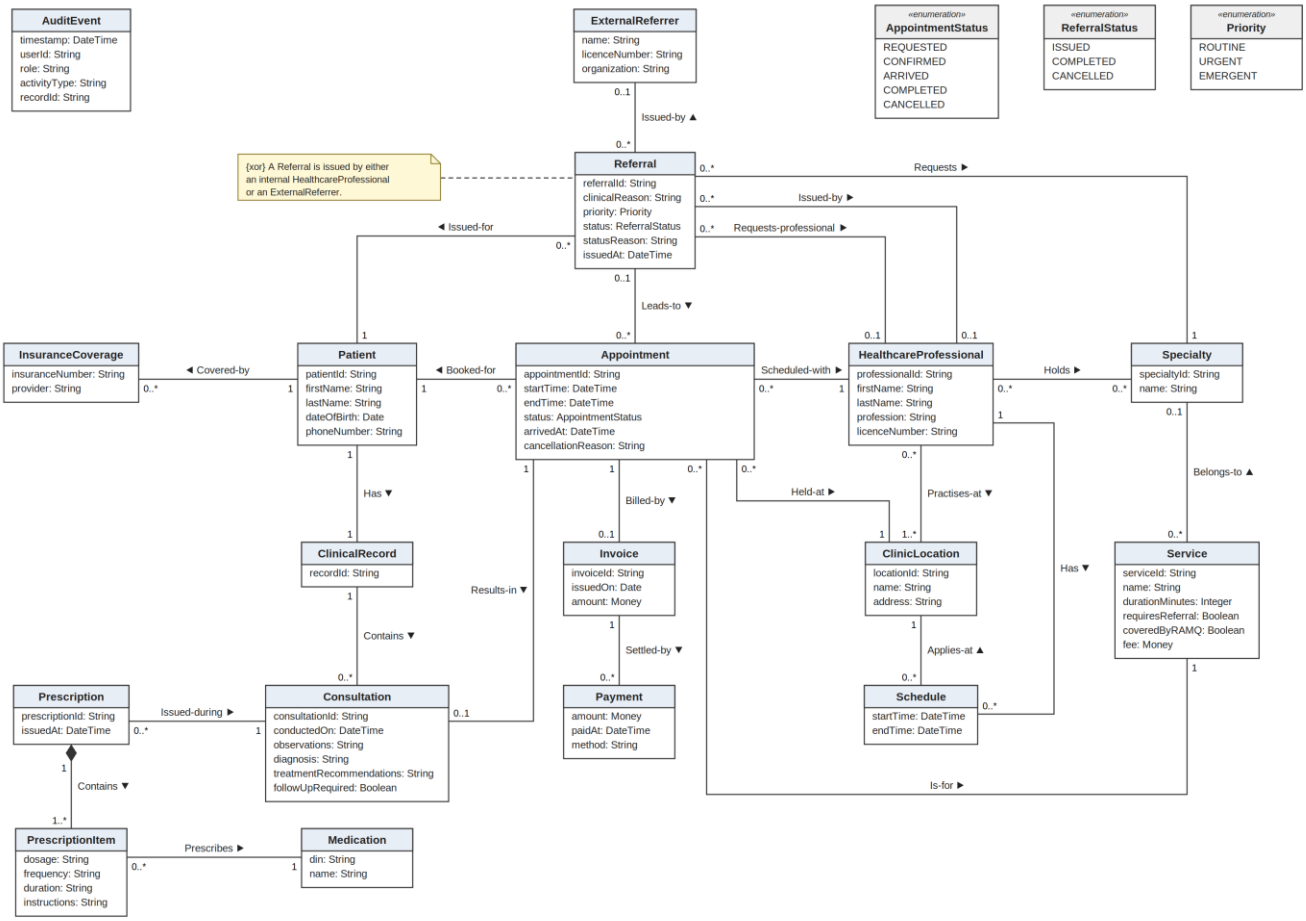
Main success scenario: 1. The professional identifies an existing referral and submits the new status (COMPLETED or CANCELLED), optionally with a reason. 2. The system verifies that the referral exists and that the status change is allowed from its current status. 3. For a cancellation, the system verifies that no confirmed appointment is still linked to the referral. 4. The system updates the referral status, records the activity, and confirms the new status.

Failure scenario: If the referral is unknown, is already COMPLETED or CANCELLED, or a cancellation is requested while a confirmed appointment is still linked, the system rejects the change and the referral status remains unchanged.

System operation: updateReferralStatus(referralId, newStatus, reason?)

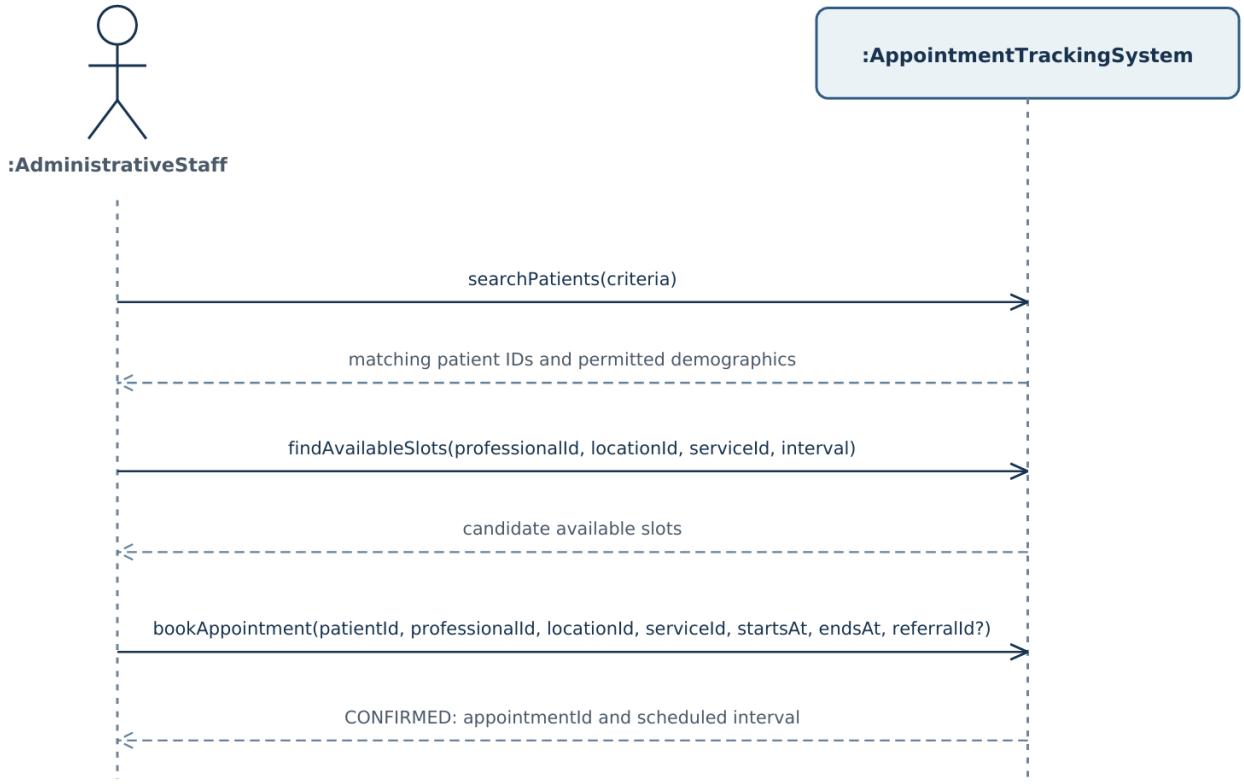
Note: Allowed changes: ISSUED to COMPLETED, and ISSUED to CANCELLED. Linked appointments must be cancelled first (N7).

UML Domain Model Diagram

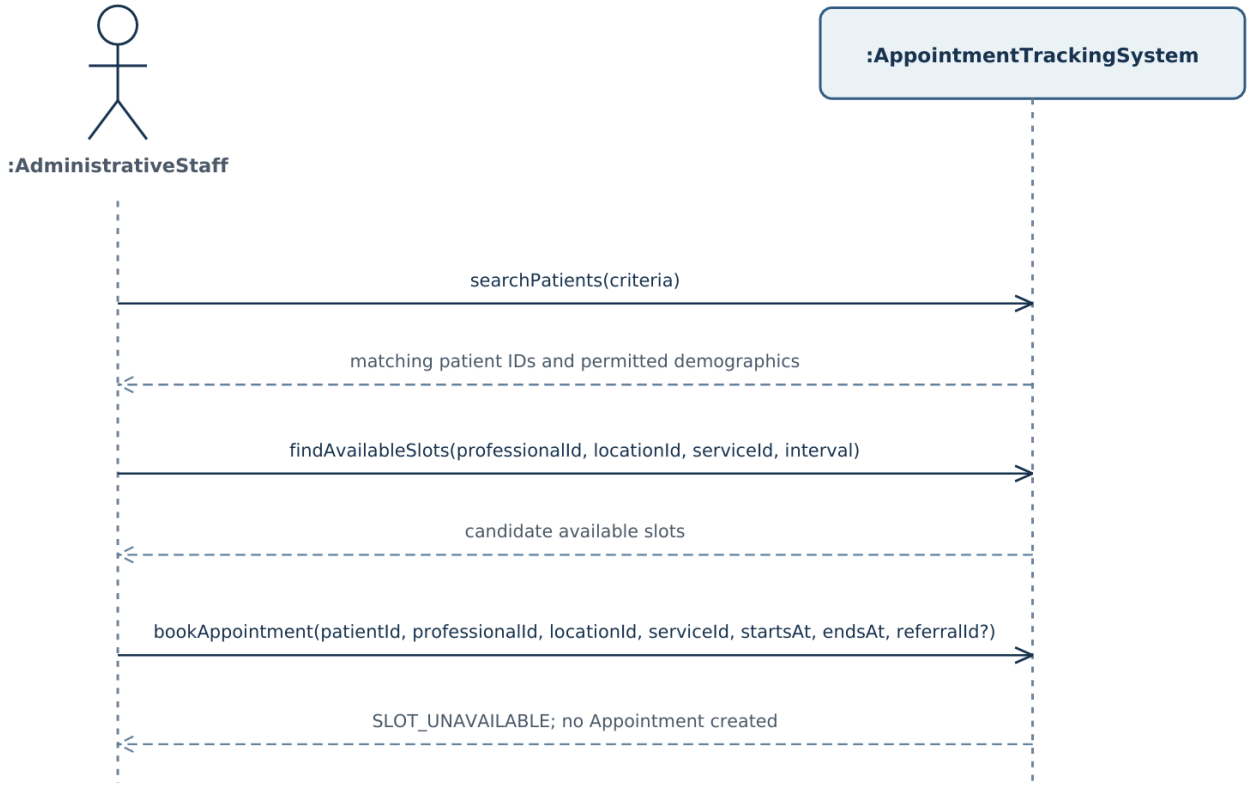


System Sequence Diagram

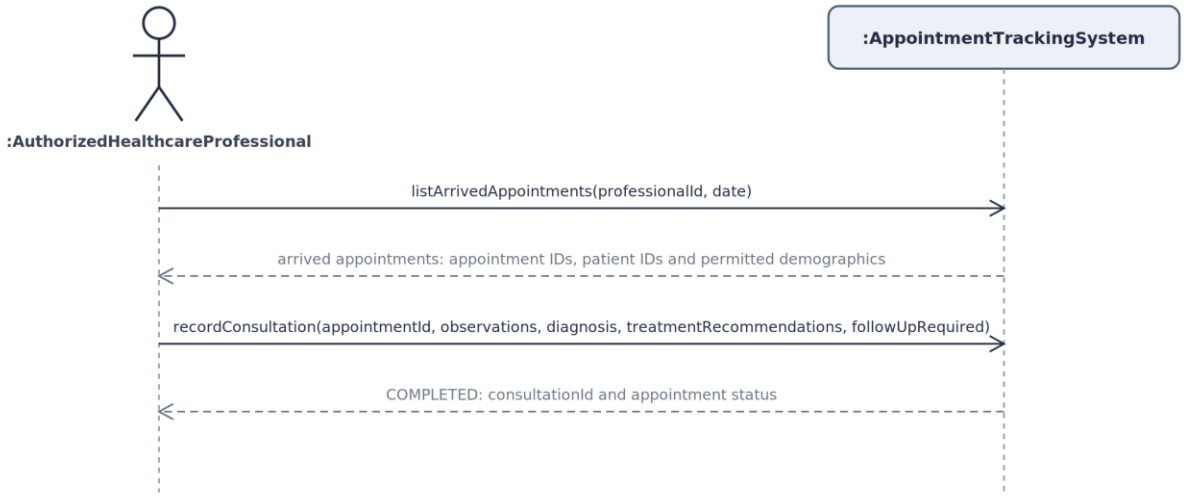
C1 – Book Appointment Success: the system revalidates availability and referral requirements before creating a confirmed booking



C1 – Book Appointment Failure: a previously offered slot is no longer available.

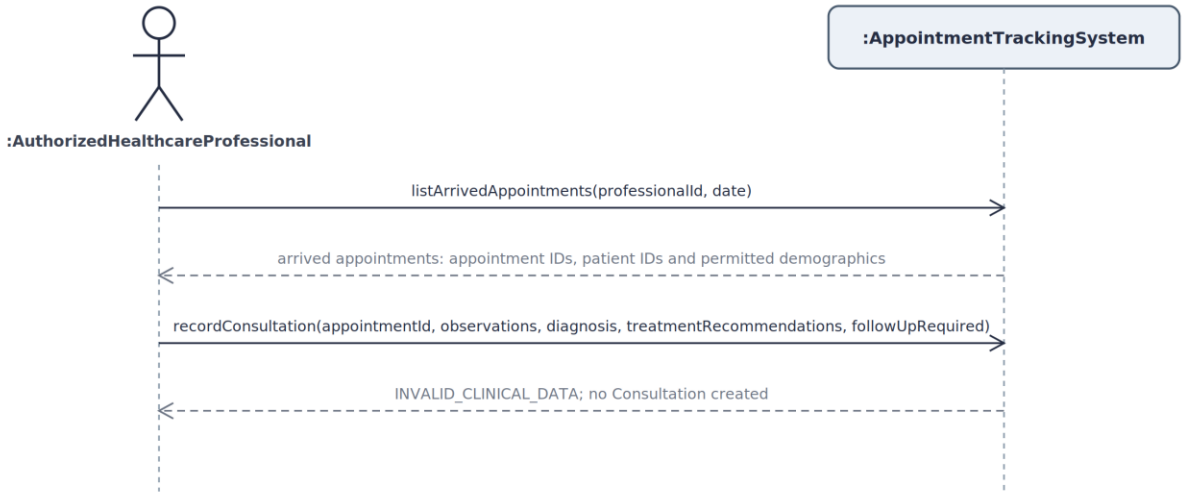


C2 - Conduct and Record Consultation Success:



The system validates the appointment and clinical data, records the consultation and completes the appointment.

C2 - Conduct and Record Consultation Failure:



Required clinical information is missing, so no Consultation is recorded.

C3 - Issue Consultation Referral Success: the system validates the referral data and creates an ISSUED referral.



C3 – Issue Consultation Referral Failure: the referral data is invalid, so no Referral is created.



Operations and Contracts

Use case	System Operation
C1	searchPatients(criteria), findAvailableSlots(professionalId, locationId, serviceId, interval), bookAppointment(patientId, professionalId, locationId, serviceId, startsAt, endsAt, referralId?)
C2	listArrivedAppointments(professionalId, date), recordConsultation(appointmentId, observations, diagnosis, treatmentRecommendations, followUpRequired)
C3	issueReferral(patientId, referrer, requestedSpecialtyId, requestedProfessionalId?, clinicalReason, priority)
N1	registerPatient(patientData)
N2	searchPatients(criteria) (same as C1)
N3	updateProfessionalSchedule(professionalId, locationId, slots)
N4	maintainService(serviceData)
N5	registerArrival(appointmentId)
N6	rescheduleAppointment(appointmentId, newStartsAt, newEndsAt)
N7	cancelAppointment(appointmentId, reason?)
N8	issuePrescription(consultationId, prescriptionItems)
N9	generateInvoice(appointmentId)
N10	registerPayment(invoiceId, amount, method)
N11	viewClinicalRecord(patientId)
N12	reviewAuditEvents(filter)

N13	confirmAppointment(appointmentId)
N14	updateReferralStatus(referralId, newStatus, reason?)

System operations from C1 – Booking Appointment system sequence diagrams:

- searchPatients(criteria)
- findAvailableSlots(professionalId, locationId, serviceId, interval)
- bookAppointment(patientId, professionalId, locationId, serviceId, startsAt, endsAt, referralId?)

Operation: searchPatients(criteria: PatientSearchCriteria)

Preconditions: The actor has permission to search patient identifiers (at least one supported search criterion is supplied).

Success postconditions: 1. The system returned the matching Patient records as permitted identifying/demographic results (possibly an empty set). 2. No Patient was created or changed by the query.

Failure / alternate outcome: If there are no matching records, an empty result is returned. Unauthorized access is rejected without disclosing patient data.

Operation: findAvailableSlots(professionalId: ProfessionalId, locationId: LocationId, serviceId: ServiceId, interval: TimeInterval)

Preconditions: The actor may view appointments/schedules and also the specified professional, location and service are identifiable.

Success postconditions: 1. The system returned zero or more available candidate time intervals subject to known scheduling constraints. 2. No Appointment was created, and no time slot was reserved by the query.

Failure / alternate outcome: If no interval qualifies, an empty set is returned. The returned slots are candidates, not a guarantee against later concurrent booking.

Operation: bookAppointment(patientId: PatientId, professionalId: ProfessionalId, locationId: LocationId, serviceId: ServiceId, startsAt: DateTime, endsAt: DateTime, referralId: ReferralId [optional])

Preconditions:

- The actor is authorized to book, and referenced Patient, HealthcareProfessional, ClinicLocation and Service exist.
- The supplied time interval is well formed.
- Availability and referral eligibility are business conditions validated inside the operation, rather than assumed beforehand.

Success postconditions:

1. A new Appointment instance a was created. [instance creation]
2. a.startTime, a.endTime and a.status were set to the accepted interval and CONFIRMED. [attribute modification]
3. a was associated with the selected Patient, HealthcareProfessional, ClinicLocation and Service. [association formation]
4. If the service required an eligible Referral, a was associated with that Referral. [association formation]
5. A relevant AuditEvent was recorded for the successful booking. [instance creation / association where applicable]
6. The confirmed appointment identifier and scheduled interval were returned to the actor. [observable output]

Failure / alternate outcome: If the selected slot conflicts with current scheduling rules, return SLOT_UNAVAILABLE. If the required referral is missing or invalid, return REFERRAL_REQUIRED or INVALID_REFERRAL. In all such cases: create no Appointment and preserve existing appointment/schedule data.

System operations from C2 - Conduct and Record Consultation system sequence diagrams:

- listArrivedAppointments(professionalId, date)
- recordConsultation(appointmentId, observations, diagnosis, treatmentRecommendations, followUpRequired)

Operation: listArrivedAppointments(professionalId: ProfessionalId, date: Date)

Preconditions: The actor is authenticated and authorized to view the appointments of the specified healthcare professional, and the specified professional is identifiable.

Success postconditions: 1. The system returned the Appointments of the specified professional on the given date whose status is ARRIVED, each with its appointment identifier, patient identifier and permitted demographic details (possibly an empty set). 2. No Appointment, Consultation or Patient was created or changed by the query.

Failure / alternate outcome: If no patient has arrived for the professional on that date, an empty set is returned. Unauthorized access is rejected with NOT_AUTHORIZED without disclosing patient data.

Operation: recordConsultation(appointmentId: AppointmentId, observations: String, diagnosis: String, treatmentRecommendations: String, followUpRequired: Boolean)

Preconditions:

- The actor is authenticated and authorized to record clinical information.
- An appointment identifier and the clinical information are supplied.
- Existence of the appointment, its assignment to the acting professional, its ARRIVED status, the absence of an existing consultation and the presence of the required clinical information are business conditions validated inside the operation.

Success postconditions:

1. A new Consultation instance c was created. [instance creation]
2. c.consultationId was assigned a unique value; c.conductedOn was set to the current date and time; c.observations, c.diagnosis, c.treatmentRecommendations and c.followUpRequired were set to the supplied values. [attribute modification]
3. c was associated with the Appointment a identified by appointmentId. [association formation]
4. c was associated with the ClinicalRecord of the Patient associated with a. [association formation]
5. a.status was changed from ARRIVED to COMPLETED. [attribute modification]
6. A relevant AuditEvent was recorded for the recorded consultation, referencing c by identifier only. [instance creation / association where applicable]
7. The consultation identifier and the appointment status COMPLETED were returned to the actor. [observable output]

Failure / alternate outcome: If the observations are missing or the follow-up indicator is not given, return INVALID_CLINICAL_DATA. If the appointment does not exist, return APPOINTMENT_NOT_FOUND. If the appointment is CANCELLED or COMPLETED, return INVALID_APPOINTMENT_STATUS. If the patient's arrival has not been registered, return PATIENT_NOT_ARRIVED. If the actor lacks authorization or the appointment is assigned to another professional, return NOT_AUTHORIZED. In all such cases: create no Consultation and leave the appointment and the patient's clinical record unchanged.

System operations from C3 - Issue Consultation Referral system sequence diagrams:

- searchPatients(criteria) (already specified under C1; reused unchanged)
- issueReferral(patientId, referrer, requestedSpecialtyId, requestedProfessionalId?, clinicalReason, priority)

Additional referral operation from N14 – Update Referral Status:

- updateReferralStatus(referralId, newStatus, reason?)

Operation: issueReferral(patientId: PatientId, referrer: ReferrerRef, requestedSpecialtyId: SpecialtyId, requestedProfessionalId: ProfessionalId [optional], clinicalReason: Text, priority: Priority)

Note: ReferrerRef is either the ProfessionalId of an internal HealthcareProfessional or the details of an external physician (name, licence number, organization).

Preconditions:

- The actor is authenticated and authorized to issue referrals
- The referenced Patients exists and the referrer is identifiable
- Specialty validity, specialty held by the professional, presence of a clinical reason and an allowed priority are conditions validated inside the operation

Success postconditions:

1. A new Referral instance r was created. [instance creation]
2. r.referralId was assigned a unique value; r.clinicalReason and r.priority were set to the supplied values; r.issuedAt was set to the current date and time; r.status was set to ISSUED. [attribute modification]
3. r was associated with the selected Patient. [association formation]

4. r was associated with the referring HealthcareProfessional; or, for an external referrer, a new ExternalReferrer instance e was created with the supplied details and r was associated with e. [instance creation / association formation]
5. r was associated with the requested Specialty and, if supplied, with the requested HealthcareProfessional. [association formation]
6. r was not associated with any Appointment. Appointments may be associated later through bookAppointment. [none]
7. A relevant AuditEvent was recorded for the issued referral. [instance creation / association where applicable]
8. The referral identifier and status ISSUED were returned to the actor. [observable output]

Failure / alt outcome: If the clinical reason is missing, the priority is not allowed, the request specialty is unknown, or the professional doesn't hold the requested specialty, return INVALID_*REFERRAL*_DATA. If actor lacks auth, return NOT_AUTHORIZED without disclosing patient data. To resume, create no Referral and preserve existing patient, referral and appointment data.

Operation: updateReferralStatus(referralId: ReferralId, newStatus: ReferralStatus, reason: Text [optional])

Preconditions:

- The actor is authenticated and authorized to update referrals.
- A referral identifier and a new status (COMPLETED or CANCELLED) are supplied.
- Existence of the referral, the validity of the status change and the absence of linked confirmed appointments (for cancellation) are business conditions validated inside the operation.

Success postconditions:

1. r.status was changed from ISSUED to newStatus. [attribute modification]
2. If a reason was supplied, r.statusReason was set to it. [attribute modification]
3. No Appointment, Patient or other Referral was created, changed or deleted. [none]
4. A relevant AuditEvent was recorded for the status change. [instance creation / association where applicable]
5. The referral identifier and its new status were returned to the actor. [observable output]

Failure / alternate outcome: If the referral does not exist, return REFERRAL_NOT_FOUND. If its status is not ISSUED, return INVALID_STATUS_TRANSITION. If CANCELLED is requested while a confirmed appointment is linked to the referral, return REFERRAL_HAS_ACTIVE_APPOINTMENTS. If the actor lacks authorization, return

NOT_AUTHORIZED. In all such cases the referral status and all other data remain unchanged.